Home Scale Carbon Calculator

Project Overview & Methodology Document
Prepared for Portfolio Documentation

1. Executive Summary

The Home Scale Carbon Calculator is an interactive Python application engineered using the Streamlit framework. It enables users to dynamically calculate their personal carbon footprint by quantifying emissions produced through daily activities. The tool leverages localized emission factors to provide a realistic assessment of environmental impact relative to national sustainability goals and the global carbon budget.

2. Mathematical Methodology & Baselines

Calculations convert user inputs (distance, power usage, waste, and meals) into annual metric tonnes of CO_2. The current computational model utilizes a dictionary of baseline multipliers optimized for the demographic and infrastructural realities of India.

Category	Factor (kg CO_2)	Justification & Source
Transportation	0.14 per km	Accounts for the average emission profile of typical passenger ICE vehicles operating within Indian infrastructural limits.
Electricity	0.68 per kWh	Aligned with Central Electricity Authority (CEA) of India data, reflecting the carbon intensity of a predominantly coal-reliant grid.
Diet	1.25 per meal	Reflects the lower carbon footprint of traditional plant-based regional diets, avoiding the high methane factors of meat-heavy diets.
Waste Management	0.10 per kg	A proxy measurement for greenhouse gases (primarily methane) released during the anaerobic decomposition of municipal solid waste in landfills.

3. Technical Architecture

The application is modular by design. The primary interface and computation engine are housed within `app.py`. Geographic flexibility is achieved via the `pycountry` library, which dynamically maps all recognized sovereign states to the UI dropdown. To ensure application stability, a dictionary comprehension technique is utilized to securely default unmapped regions to the established baseline factors until localized research is integrated.

4. Strategic Insights & Context

Rather than merely outputting raw data, the software utilizes conditional logic to furnish contextual environmental warnings based on the chosen jurisdiction. For example, selecting India prompts an alert contextualizing the user's footprint against the projected 2025 per capita national average (2.18 tonnes), aligning the individual metric with broader UNEP warming limit guidelines.